Grass and Forage Science
The Journal of the British Grassland Society | The Official Journal of the European Grassland Federation

Effects of species richness and functional groups on chemical constituents relevant for methane yields from anaerobic digestion: results from a grassland diversity experiment

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Abstract

Changes in livestock production systems have led to land-use changes and abandonment, especially of semi-natural grassland in agriculturally less favoured regions. The generation of energy from biomass of extensive, high-diversity grasslands can be an alternative to their abandonment, and anaerobic digestion is one possible method for converting grassland biomass into energy. However, little is known about the effects of species richness (SR) and functional groups on chemical constituents relevant for anaerobic digestion and the resulting energy potential. In this study, changes in the herbage chemical constituents that are relevant for forage quality were studied along a well-defined diversity gradient (one to sixty species) and across different combinations of functional groups (legumes, small herbs, tall herbs and grasses). Substrate-specific methane yield ($CH_{4\ sub}$) was estimated through the concentrations of forage-quality parameters such as crude fibre (CF), crude protein (CP), crude lipid, nitrogen-free extract and their documented digestibility values, as well as the respective methane yields. Results show that with increasing SR, the CF increased and CP decreased, even though these effects could not be fully disentangled from the presence of grasses. These trends led to a negative effect of SR on $CH_{4\ sub}$, while the area-specific methane yield ($CH_{4\ area} = CH_{4\ sub} \times$ biomass yield) increased due to a strong increase in biomass with increasing SR. The $CH_{4\ sub}$ was increased when legumes were present, and it declined with the presence of grasses. Generally, $CH_{4\ sub}$ and $CH_{4\ area}$ varied between functional-group monocultures and all functional-group mixtures.

Keywords: digestibility, diversity, bioenergy, fibre, functional groups, methane

Introduction

Semi-natural grasslands, originally used to supply forage for livestock production, also provide many ecosystem functions and are often important hotspots of biodiversity in rural areas. However, their species diversity often declines when the land-use changes, whether as a result of intensification or abandonment (Mitchley, 2001). In countries that have developed productive systems of livestock agriculture, the feed quality of forage from semi-natural grassland is generally inadequate to meet the nutritional requirements for high levels of milk output or rapid liveweight gain, and this has resulted in many grassland areas being abandoned by farmers (Isselstein *et al.*, 2005). For some regions in Germany, it has been estimated that up to one quarter of the permanent grassland area is likely to be abandoned in the near future (Rösch *et al.*, 2007). This is leading to a surplus of permanent grassland that requires regular management through grazing and/or cutting if its species richness (SR) is to be protected (Mitchley, 2001).

Grassland biomass that is no longer required to supply forage for animal production can be used

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Received 14 May 2012; revised 30 October 2012

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for energy production without necessarily creating competition for food production and possible environmental drawbacks. Anaerobic digestion of extensive grassland biomass for biogas production is one conversion technique that can be used in the management of grassland. In providing a use for surplus grassland, it can contribute to maintaining biodiversity and, at the same time, create an economic incentive for farmers. Overall, in developed countries, the production of biogas from surplus grassland has a high potential for bioenergy (Prochnow *et al.*, 2009). On a European scale, 10·9 million tons of oil equivalents (Mtoe) were produced as biogas in the year 2010, which was a 31·3% increase compared with the previous year (EurObserv'ER, 2011). Among the substrates used for biogas production in Germany, grass silage is one of the most commonly used co-substrates (used in 40% of the biogas plants in Germany) after maize (which is used in 90% of all plants; Weiland, 2006; FNR, 2009). However, the percentage of extensive grassland contributing to 'grass silage' remains unspecified.

In Prochnow *et al.* (2009), more than fifteen European studies on biogas and methane yields from different types of grassland, grown under different abiotic conditions and with different management strategies, were summarized and compared. However, the subject of how SR or functional-group composition in extensive grassland may be linked directly with biogas potential has so far not been addressed. For farmers managing extensive grasslands within the framework of nature conservation programmes (e.g. Natura 2000), it can be important to understand how changes in botanical diversity may affect the potential methane yield. This determines the associated economic value of the site beyond that provided by environmental support payments for nature conservation.

In a multi-site study with managed grassland, White *et al.* (2004) showed the relationship between measures of vegetation quality [such as crude protein (CP) and fibre content (measured as neutral detergent fibre)] and SR. In a summer and autumn cut, CP concentrations were lower in high-SR sites than in low-SR sites, while the fibre tended to be higher only in herbage at the second cut of high-SR sites. However, the results presented by White *et al.* (2004) show differences in biomass quality with different levels of SR as a result of site-dependent factors such as resource availability and therefore do not indicate SR effects *per se*. Yet, combining the observed relations between SR and concentrations of crude fibre (CF) and CP with what is known from research on methane yields of carbohydrates (including CF), CP and crude lipid (CL; Baserga, 1998), and from a study indicating a decline in plant-available N in the soil solution with increasing SR (Oelmann *et al.*, 2007), it can be expected that the substrate-specific methane yield ($CH_{4\ sub}$) would decline along a gradient of increasing SR. The $CH_{4\ sub}$ expresses the potential energy contained in the biomass that can be realized through anaerobic digestion, and it can be used as a measure for comparing the biogas potentials of different substrates. In addition to the potential effects of SR on $CH_{4\ sub}$, there is a lack of knowledge on the variation in $CH_{4\ sub}$ of grassland biomass caused by the differences in plant species composition (Prochnow *et al.*, 2009).

If $CH_{4\ sub}$ declines with SR, the question arises of how the area-specific methane yield ($CH_{4\ area}$) will be affected by the negative SR–$CH_{4\ sub}$ relationship? In general, area-specific energy yield (E_{area}) is calculated as the product of biomass yield and substrate-specific energy yield (E_{sub}) and is important for assessing the potential value of a grassland site for bioenergy production. In a study on solid-fuel quality of grassland biomass for combustion and the effect of SR on E_{area}, it was shown that E_{area} increased with SR due to a biomass yield increase, while E_{sub} remained constant (Khalsa *et al.*, 2012).

Thus, the aim of this study was to approach systematically the relationships of functional groups (presence/absence or abundance), as representatives of plant species with similar functional traits, and SR with (i) chemical constituents relevant for methane yield from anaerobic digestion, (ii) $CH_{4\ sub}$ and (iii) $CH_{4\ area}$ and annual $CH_{4\ area}$.

The herbage biomass investigated in this study came from a large-scale biodiversity grassland experiment with artificial species mixtures set up along two diversity gradients: SR and functional-group richness (FGR; Roscher *et al.*, 2004). Testing of SR effects was given special attention in the design of the experiment, to disentangle as much as possible the effects of SR, FGR and the presence of individual functional groups. To address criticisms provoked by previous diversity studies, the choice of functional groups was also considered very carefully by giving special attention to forbs and legume species. Legumes were therefore regarded as a separate functional group, as they have been found to have disproportionate effects on ecosystem processes (Spehn *et al.*, 2002). Whereas previous biodiversity experiments have usually included three functional groups [grasses, forbs ('herbs') and legumes], it was decided in this experiment to further divide the 'herbs' group into small herbs and tall herbs.

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Materials and methods

Experimental design

In May 2002, an experimental site with semi-natural mesophilic grassland was established on the floodplain of the River Saale (near Jena, Thuringia, Germany; 50°55N, 11°35E, 130 m a.s.l.). Mean annual air temperature in the Jena area is 9·3°C, and annual precipitation is 587 mm (Kluge *et al.*, 2000). The site had originally been grassland and was converted into arable land around 1960. Soil conditions resemble a Eutric Fluvisol (FAO, 1994), and soil texture changes from silty clay to sandy loam with increasing distance from the river.

Sixty plant species were used to create a gradient in plant SR (1, 2, 4, 8, 16 and 60) and in FGR (1, 2, 3 and 4). Functional groups were defined, according to the morphological, phenological and physiological traits of the plant species, as grasses (n = 16), small herbs (n = 12), tall herbs (n = 20) and legumes (n = 12; a detailed list of plant species can be found in Roscher *et al.*, 2004). Eighty-two plots (20 × 20 m) were established in four blocks, the blocks accounting for the differences in soil texture. Sixteen possible combinations of SR and FGR were realized and replicated over the four blocks (Figure 1). The location of the mixtures within each block was fully randomized. The management of the site consisted of two cuts per year (late May and late August) with no fertilizer applications; this management was considered typical for semi-natural grassland under the actual site conditions. Plots were weeded twice a year to maintain the original species composition. A full description of the experimental setup was published in Roscher *et al.* (2004).

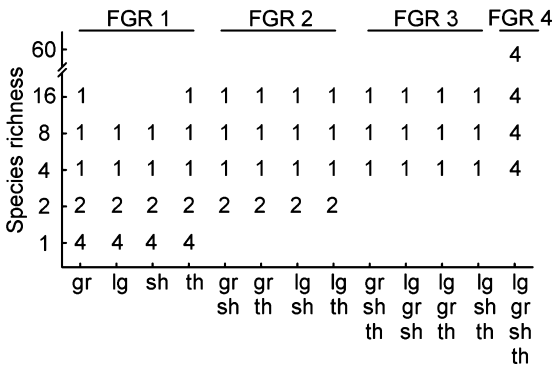

Figure 1 Combinations of functional groups and their replications (numbers) according to species richness level. Each replication is representing one of the plots used in this study; the total sum therefore is 82. The *x*-axis gives the different functional-group compositions (gr, grasses; lg, legumes; th, tall herbs and sh, small herbs). Functional-group richness (FGR) is shown on top.

Biomass sampling

Above-ground biomass was harvested twice in 2008 and 2009, in late May and late August (immediately before cutting). Three randomly placed samples of 20 × 50 cm were harvested at 3 cm above the soil surface. Biomass samples were separated into target species, dead plant material and non-target species, dried (70°C, 48 h) and weighed. Mean total biomass (t DM ha^{-1}) was derived from the three samples. Annual biomass was calculated as the sum of biomass from the first and second cut. An additional 300 g of above-ground biomass was sampled from each plot (n = 82) in both years and both cuts (total n = 164) for forage-quality analysis.

Forage-quality parameters

From each of the 164 biomass samples, the following were determined: crude ash (CA), by loss on ignition at 550°C, and CP, calculated from the nitrogen (N) content (CP = 6·25 × N) analysed with an elemental analyser (vario MAX CHN; Elementar Analysensysteme GmbH, Hanau, Germany).

For CF and CL, 100 of the 164 samples were analysed in the laboratory according to standard methods (Naumann and Bassler, 2004) and then used for near-infrared calibration. A near-infrared spectroscope (XDS Rapid Content Analyser; FOSS NIRSystems Inc., Laurel, MD, USA) was used, and all values for the 164 samples were predicted after cross-validation (CF: R^2 = 0·96, RPD = 3·68; CL: R^2 = 0·74, RPD = 1·70).

Nitrogen-free extract (NfE) was calculated as: NfE (% DM) = 100 − CA (% DM) − CF (% DM) − CL (% DM) − CP (% DM), and the sum of NfE and CF was assumed to represent the fraction of carbohydrates.

Estimating substrate-specific methane yield

To calculate $CH_{4\ sub}$ for this heterogeneous data set, the guidelines of VDI 4630 (VDI, 2006) were used and modified to also account for the effects of botanical composition and date of harvest in the calculations. The method was based on the concentrations of the constituents CF, CL and CP (g kg^{-1} DM), a digestibility coefficient (D) for each constituent and the potential biogas yield (B) of carbohydrates, CL and CP. The B values (which were estimated through stoichiometric equations) and the average methane content of the biogas (M) produced by the different constituents were taken from VDI 4630 (VDI, 2006; Table 1). $CH_{4\ sub}$ is expressed in norm litres

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Table 1 Maximum biogas production (*B*) from different chemical constituents [crude protein (CP), crude lipid (CL), carbohydrates] and average methane content (*M*) in the biogas according to VDI 4630 (2004).

Constituent	B (l kg^{-1} VS)	M (%/100)
Carbohydrates (CF + NfE)	746	0·50
CL	1390	0·72
CP	800	0·70

per kg volatile solids (VS), which refer to norm conditions of 273·2 K and 101·3 kPa:

$$\begin{aligned} CH_{4\,sub}(l_n\ kg^{-1}\ VS) = &(CF \times D_{CF} + NfE \times D_{NFE}) \\ &\times B_{(CF+NfE)} \times M_{(CF+NfE)} \\ &+ CL \times D_{CL} \times B_{CL} \times M_{CL} \\ &+ CP \times D_{CP} \times B_{CP} \times M_{CP} \end{aligned}$$

In this study, D was calculated for each plot individually with respect to its dominant functional group, date of cut and the CF content of the total biomass. This was accomplished by linear regressions based on a data set by the University of Hohenheim (2007) (Table 2), which contains mean values derived from investigations ranging from 84 to 557 samples. The linear regressions were based on $n = 7$ values for the grass-dominated plots and $n = 7$ values for the legume-dominated and herb-dominated plots.

Statistical analysis

The Jena Experiment was designed to vary SR, FGR and functional-group composition (FGC) as orthogonally as possible (Figure 1). However, a fully balanced design is not possible; for instance, the lowest SR cannot be combined with the highest FGR. This is not an unusual situation in biodiversity experiments and can be approached by analysing the dependent variable in an analysis of variance (ANOVA) with sequential sum of squares (Schmid *et al.*, 2007). In this type of analysis, variables that are fitted before others take up all the variation they can explain, ignoring the possibility that the later variables might also explain some of this variation (McCullagh and Nelder, 1999; Hector *et al.*, 2010). The characteristics of this type of analysis can then be used to identify effects that are independent of the variables fitted previously.

To account for the gradient in soil conditions, block-wise weeding and mowing, as well as sampling, the block effects were fitted first. It can therefore be assumed that all the variance that is explained by variables fitted after the block effect is independent of it.

As the main focus of this study was on the effects of increasing SR, this variable was fitted first after block. To test the effects of SR, the log-linear contrast of one to sixteen species was used. The sixty species plots were used as a point of reference for the highest possible level of species diversity but were not included in the statistical analysis. Two plots had to be omitted from the data set (one small-herb monoculture plot and one tall-herb monoculture plot) due to difficulties with their establishment.

Table 2 Linear regressions for grassland biomass to estimate digestibility values (D, in g kg^{-1} DM) of chemical constituents [crude protein (CP), crude lipid (CL), crude fibre (CF) and nitrogen-free extract (NfE)] according to functional-group dominance (FGD), CF content (g kg^{-1} DM) and cut.

FGD	Constituent	Linear regression	R^2	RSE
Grass (>70% of biomass), 1st cut	CF	$D_{CF} = -0{\cdot}98 \times CF + 1023{\cdot}4$	0·88	2·11
	NfE	$D_{NfE} = -1{\cdot}27 \times CF + 1078{\cdot}2$	0·94	1·89
	CL	$D_{CL} = 0{\cdot}26 \times CF + 577{\cdot}1$	0·80	0·78
	CP	$D_{CP} = -1{\cdot}07 \times CF + 964{\cdot}8$	0·95	1·48
Grass (>70% of biomass), 2nd and further cuts	CF	$D_{CF} = -0{\cdot}54 \times CF + 858{\cdot}6$	0·99	0·18
	NfE	$D_{NfE} = -0{\cdot}52 \times CF + 830{\cdot}2$	0·69	1·81
	CL	$D_{CL} = -0{\cdot}99 \times CF + 863{\cdot}7$	0·70	3·36
	CP	$D_{CP} = -0{\cdot}85 \times CF + 894{\cdot}5$	0·69	3·00
Legumes + herbs (>30% of biomass), 1st cut	CF	$D_{CF} = -1{\cdot}75 \times CF + 1236{\cdot}7$	0·95	2·03
	NfE	$D_{NfE} = -2{\cdot}42 \times CF + 1361{\cdot}6$	0·98	1·80
	CL	$D_{CL} = -0{\cdot}14 \times CF + 619{\cdot}1$	0·01	6·47
	CP	$D_{CP} = -0{\cdot}87 \times CF + 938{\cdot}3$	0·91	1·33
Legumes + herbs (>30% of biomass), 2nd cut	CF	$D_{CF} = -1{\cdot}21 \times CF + 1010{\cdot}1$	0·99	0·30
	NfE	$D_{NfE} = -1{\cdot}67 \times CF + 1133{\cdot}3$	0·99	0·01
	CL	$D_{CL} = -1{\cdot}02 \times CF + 734{\cdot}0$	0·59	3·93
	CP	$D_{CP} = -0{\cdot}92 \times CF + 936{\cdot}5$	0·98	0·59

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To test for the effects of FGR and FGC, those two variables were fitted after SR in a first model. This initial model was performed as a repeated-measure ANOVA to test for the effects of the two harvest dates. In the following steps, one for each cut, the presence/absence of functional groups was fitted after SR to test for their individual effects. To avoid further complexity of the models and thus possible problems with interpretation of the underlying mechanisms (e.g. weather conditions or patterns in plant community dynamics), it was decided that the sampling year should not be included as a factor and instead to use mean values of both years for the ANOVA.

Multiple regression analysis was conducted (on all plots including SR = 60) to estimate the influence of functional-group abundance on forage quality and $CH_{4\ sub}$ by selecting the terms for inclusion in the model depending on standard statistical model selection methods (Draper and Smith, 1998). This implies that effect terms with $P < 0{\cdot}05$ were included according to the rules of hierarchy and marginality (Nelder, 1994; Nelder and Lane, 1995). The marginality principle (Nelder and Lane, 1995) requires a term that appears as part of a more complex element in the model not to be tested for significance of its main effect, because the meaning of such terms is open to misinterpretation (Connolly and Wachendorf, 2001).

The initial model contained all functional-group abundance terms (lg, gr, sh, th) including their quadratic terms and all possible pairwise interactions. Functional-group abundances were tested for co-linearity (with *cor(x, y, method = 'Pearson')* in R) prior to the analysis. The correlations between functional-group abundances were all within $-0{\cdot}21 > r < -0{\cdot}44$ and on average at $r = -0{\cdot}33$ in both cuts.

All statistical analyses were performed in R 2.14.1 (R Development Core Team, 2011).

Results

Forage quality with changing diversity

Crude fibre ranged from 157 to 369 g kg^{-1} DM across all plant species compositions and the two harvest dates, with an overall mean of 251 g kg^{-1} DM. Differences between the two cutting dates were observed, with CF content being higher in the first than in the second cut. Statistical analysis revealed that the CF content was positively affected by SR in both cuts (Table 3). As SR increased from one to sixty species, CF increased in the first cut by 65 g kg^{-1} DM and in the second cut by 34 g kg^{-1} DM (Figure 2a and b). Furthermore, grasses had a strong effect in terms of increasing the CF in both cuts, while the effect of legumes decreased the CF (Table 3). This is also reflected in the CF mean values of the functional-group monocultures, which were highest for grasses (Table 4). Predicting CF content from the functional-group abundance based on multiple regression analysis revealed that, in the first cut, few parameters (grass and small-herb abundance) were necessary to achieve a model accuracy of $R^2 = 0{\cdot}55$ (Table 5), whereas in the second cut, even with the full information of legume, grass and tall-herb abundance, the accuracy was much lower ($R^2 = 0{\cdot}35$). Furthermore, multiple regression analysis revealed an interaction between grasses and small-herb abundance in the first cut (Figure 3a), while no interactions were detected in the second cut (Figure 3b).

Crude protein, which was included as it is a major determinant of the $CH_{4\ sub}$ (even though it was previously published as nitrogen, N = CP/6·25) in Khalsa *et al.* (2012)), ranged from 60 to 228 g kg^{-1} DM with a mean of 115 g kg^{-1} DM. Differences between the two cuts were small, so that similar effects were found in the statistical analysis (Table 3). The content of CP was negatively affected by SR and declined from 121 to 90 g kg^{-1} DM in the first cut along the gradient of one to sixty species and from 129 to 103 g kg^{-1} DM in the second cut respectively (Figure 2c and d). Legumes increased the CP content, while grasses reduced it, which was again reflected in the mean values of the functional-group monocultures being highest for legumes and lowest for grasses (Table 4). The CP content could be accurately predicted ($R^2 = 0{\cdot}60$) by legume, tall-herb and small-herb abundance in the first cut (Table 5) including an interaction between legume and tall-herb abundance (Figure 3c). In the second cut, legume abundance was sufficient to predict CP content with a $R^2 = 0{\cdot}64$ (Figure 3d).

The values of CL were much lower than those of CF and CP and ranged between 11 and 36 g kg^{-1} DM, with a mean of 21 g kg^{-1} DM. The CL content was higher in the second cut than in the first, while statistical analysis revealed only few but non-consistent functional-group effects. Legumes, for example, were found to have an increasing effect in the first cut and a decreasing effect in the second cut, while grasses only had an increasing effect in the first cut (Table 3). The mean CL values in the functional-group monocultures in the first cut were relatively low in grasses compared with the other monocultures. In the second cut, grasses had the highest CL content relative to the other functional-group monocultures (Table 4).

NfE ranged from 431 to 599 g kg^{-1} DM with a mean of 519 g kg^{-1} DM. The NfE means were not significantly different between the two cuts and, as with CL, statistical analysis revealed only few but non-consistent functional-group effects between the cuts. Small herbs increased the NfE in the first cut, while in the second cut, it was decreased by legumes and

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Table 3 Analysis of variance of crude fibre (CF), crude lipid (CL), crude protein [CP, previously published as nitrogen in Khalsa *et al.* (2012)], and nitrogen-free extract (NfE) as functions of diversity parameters (species richness (SR), functional-group richness (FGR), functional-group composition (FGC) and presence/absence of individual functional groups. (A) The overall effects of diversity and the significance of cut were tested. The effects of individual functional groups were tested separately for (B) 1st cut and (C) 2nd cut. Arrows indicate increase (↑) or decrease (↓) with presence of respective functional group or with increasing SR.

		CF (g kg^{-1} DM)		CL (g kg^{-1} DM)		CP (g kg^{-1} DM)		NfE (g kg^{-1} DM)	
Factor	**DF**	***F***	***P***	***F***	***P***	***F***	***P***	***F***	***P***
(A) Block	3	1	0·244	3	0·044	<1	0·781	1	0·325
Log (SR)	1	17	<0·001↑	3	0·093	14	<0·001↓	<1	0·983
FGR	1	<1	0·852	1	0·233	<1	0·679	<1	0·822
FGC	11	9	<0·001	<1	0·997	15	<0·001	2	0·152
Residuals	59								
Repeated measures									
Cut	1	55	<0·001	66	<0·001	11	0·002	3	0·119
Cut × Log (SR)	1	2	0·225	1	0·413	<1	0·535	6	0·015
Cut × FGR	1	<1	0·677	1	0·422	3	0·109	2	0·132
Cut × FGC	11	2	0·018	6	<0·001	3	0·003	1	0·384
Residuals	62								
(B) 1st cut									
Block	3	3	0·027	1	0·258	2	0·073	1	0·274
Log (SR)	1	12	0·001↑	1	0·407	5	0·037↓	4	0·050
Legumes	1	10	0·003↓	9	0·004↑	20	<0·001↑	<1	0·728
Grasses	1	30	<0·001↑	21	<0·001↓	36	<0·001↓	<1	0·964
Tall herbs	1	<1	0·512	1	0·403	1	0·277	1	0·386
Small herbs	1	11	0·001↓	1	0·459	1	0·366	6	0·022↑
FGC	8	3	0·009	1	0·239	5	<0·001	1	0·562
Residuals	59				0·258				
(C) 2nd cut									
Block	3	3	0·032	7	<0·001	2	0·108	1	0·298
Log (SR)	1	6	0·022↑	3	0·098	13	0·001↓	2	0·143
Legumes	1	8	0·007↓	6	0·022↓	92	<0·001↑	3	0·094↓
Grasses	1	17	<0·001↑	4	0·066	9	0·005↓	3	0·119
Tall herbs	1	9	0·004↓	<1	0·511	2	0·213	5	0·023↑
Small herbs	1	1	0·312	3	0·067	11	0·001↓	<1	0·657
FGC	8	1	0·220	1	0·715	2	0·033	1	0·804
Residuals	59								

increased by tall herbs (Table 3). Regarding the functional-group monocultures, in both cuts, the legumes had the lowest NfE values, whereas small herbs were highest in the first cut and tall herbs were highest in the second cut (Table 4).

Methane yields with changing diversity

$CH_{4\ sub}$ yields of 234–387 l_n kg^{-1} VS, with an overall mean of 299 l_n kg^{-1} VS, could be estimated across all plant species compositions and the two harvest dates. Statistical analysis revealed no significant difference between the two cuts and showed similar functional-group effects (Table 6). In both cuts, SR had a decreasing effect on $CH_{4\ sub}$, which was more pronounced in the first cut than in the second cut (decrease from one to sixty species by 55 l_n kg^{-1} VS and by 26 l_n kg^{-1} VS respectively; Figure 4). Presence of legumes in both cuts, as well as small herbs in the first and tall herbs in the second cut, increased the $CH_{4\ sub}$. Presence of grasses, on the other hand, had a strong decreasing effect on $CH_{4\ sub}$ irrespective of the cut. This is also reflected in the functional-group monocultures, where grasses had the lowest mean values in both cuts, while legumes had the highest (Table 7). $CH_{4\ sub}$ could be predicted by legume, small-herb and grass abundance in the first cut (R^2 = 0·53), whereas in the second cut, the model consisted of legumes, small herbs and tall herbs (R^2 = 0·40; Table 5). The multiple regression models included interactions between grasses and

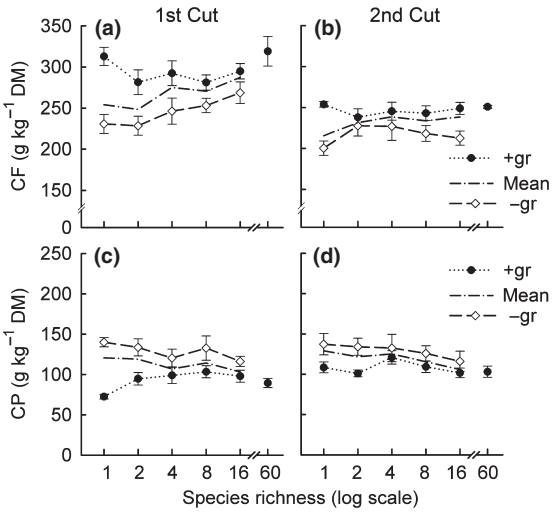

Figure 2 (a, b) Crude fibre (CF) and (c, d) crude protein (CP) content with increasing species richness plotted separately for each cut. Data for grasses present (+gr: solid circles and dotted line) and grasses absent (−gr: open diamonds and dashed line) were plotted individually. Overall trend of mean values with increasing species richness was indicated by dash-dotted line. Values are means of both years. Error bars indicate standard error. Graphs (c) and (d) were previously published as nitrogen in Khalsa *et al.* (2012).

small herbs in the first cut (Figure 5a) and between tall herbs and small herbs in the second cut (Figure 5b).

$CH_{4\ area}$ had a mean value of 656 m^3 ha^{-1} with a wide range from 131 to 1654 m^3 ha^{-1}, which was largely due to the variance in biomass yield of 0·5–7·6 t DM ha^{-1}. As the biomass yield between the two cuts was significantly different, with 3·2 t DM ha^{-1} in the first cut and 1·4 t DM ha^{-1} in the second cut, the $CH_{4\ area}$ values were 867 m^3 ha^{-1} and 445 m^3 ha^{-1} respectively. Due to the close relationship between biomass yield and $CH_{4\ area}$ (R^2 = 0·97), the effects found in the statistical analysis were very similar (Table 6). There was a positive effect of SR in both cuts, with $CH_{4\ area}$ increasing from one to sixty species by 752 m^3 ha^{-1} and by 296 m^3 ha^{-1}, in the first and second cuts respectively (Figure 6). Legume presence increased $CH_{4\ area}$ in both cuts, and this was also reflected in the mean values of the functional-group monocultures, where legumes had the highest $CH_{4\ area}$ value in the second cut and the second-highest value in the first cut (Table 7).

The annual $CH_{4\ area}$ also had a wide range (378–2595 m^3 $ha^{-1}a^{-1}$) with a mean of 1292 m^3 ha^{-1} a^{-1}. It was significantly affected by SR (P < 0·001) from one to sixty species, and it increased by more than 100% from 979 to 2027 m^3 ha^{-1} a^{-1} (Figure 6). The presence of legumes had a strong effect (P < 0·001) on annual $CH_{4\ area}$, which was in line with the patterns found in the functional-group monocultures where legumes had the highest annual $CH_{4\ area}$ (Table 7).

Discussion

Chemical constituents affecting anaerobic digestion and $CH_{4\ sub}$

Biogas production is influenced by many factors such as temperature in the digestion plant, pH value of the digestate, diversity of microorganisms and concentration of trace elements (Weiland, 2010). Furthermore, the actual methane yield is influenced by the retention time, type of digestion system and substrate quality of the feed. An ideal substrate for biogas production should be rich in degradable carbohydrates (e.g. sugars), lipids and proteins and at the same time be poor in hemicelluloses and lignin (El Bassam, 1998).

In general, high fibre contents limit energy availability from forages (Buxton and Redfearn, 1997) and, in particular, high lignin contents are not suitable for biogas production (Zubr, 1986). Chopping of the harvested biomass, as for instance in the preparation of material for ensiling, helps to break down the lignocellulose structures in the fermenter and may lead to a better digestibility of the fibre fraction (Prochnow *et al.*, 2009).

High protein concentrations, on the other hand, may lead to process failure due to an accumulation of ammonia, whereas a balanced C/N ratio of 25 is expected to gain the highest methane yields (Zubr, 1986). Calculated from the data presented in Khalsa *et al.* (2012), the C/N ratio of the all functional-group mixtures (first cut: 27·6; second cut: 24·5) was near the optimum.

A consideration of forage quality in terms of its quality requirements for ruminant nutrition may provide a first simple approach to estimate whether a substrate is valuable for energy production via anaerobic digestion. For dairy production, the forage quality of grassland biomass is regarded as low at CF values of 350–400 g kg^{-1} DM and CP values of approximately 63 g kg^{-1} DM, while CF between 150 and 200 g kg^{-1} DM and CP of 180–220 g kg^{-1} DM would indicate high quality (Peeters, 2009). However, anaerobic digestion in the digestive tracts of ruminants may differ from anaerobic digestion in a biogas plant. Biogas plants can degrade cellulose to a much greater extent than ruminants, partly because the digestion time in biogas plants is longer and the

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Table 4 Mean values including standard deviation (s.d.) of crude fibre (CF), crude protein (CP), crude lipid (CL) and nitrogen-free extract (NfE) in functional-group (FG) monocultures as well as the all functional-group mixtures. Values are means of both years.

	Grasses		Legumes		Small herbs		Tall herbs		All FG mixtures	
Constituent	**Mean**	**s.d.**	**Mean**	**s.d.**	**Mean**	**s.d.**	**Mean**	**s.d.**	**Mean**	**s.d.**
CF (g kg^{-1} DM)										
1st cut	321	±18·2	235	±37·2	218	±24·7	258	±35·7	283	±36·2
2nd cut	262	±9·30	208	±23·7	229	±34·7	200	±22·7	237	±21·7
CP (g kg^{-1} DM)										
1st cut	71·7	±6·11	162	±28·5	135	±12·9	115	±17·3	104	±22·4
2nd cut	99·4	±13·3	175	±24·9	106	±21·9	108	±9·81	112	±21·3
CL (g kg^{-1} DM)										
1st cut	16·4	±2·63	20·8	±3·00	21·2	±1·39	20·2	±2·32	19·1	±2·23
2nd cut	29·3	±3·70	23·0	±3·90	22·5	±6·02	22·8	±2·71	22·6	±3·91
NfE (g kg^{-1} DM)										
1st cut	523	±17·9	509	±18·7	539	±24·0	521	±33·6	517	±18·6
2nd cut	506	±16·6	503	±29·1	522	±19·0	545	±32·0	524	±16·2

Table 5 Parameter estimates for the multiple regression analysis of crude fibre (CF), crude protein (CP) and substrate-specific methane yield ($CH_{4\ sub}$). Abundances for legumes (lg), grasses (gr), tall (th) and small herbs (sh) are given as DM yield percentage.

CF (g kg^{-1} DM)			CP (g kg^{-1} DM)			$CH_{4\ sub}$ (l_n kg^{-1} VS)		
Effect	**Estimate**	***P***	**Effect**	**Estimate**	***P***	**Effect**	**Estimate**	***P***
1st cut								
Intercept	257·167	<0·001	Intercept	75·998	<0·001	Intercept	293·674	<0·001
gr	0·609	<0·001	lg	0·810	<0·001	lg	0·222	0·021
sh	−0·432	<0·001	th	0·375	<0·001	gr	−0·225	0·012
gr × sh	0·017	0·005	sh	0·493	<0·001	sh	0·437	<0·001
			lg × th	−0·012	0·016	gr × sh	−0·017	<0·001
R^2 = 0·55	RSE = 30·01		R^2 = 0·60	RSE = 19·90		R^2 = 0·53	RSE = 22·21	
2nd cut								
Intercept	243·110	<0·001	Intercept	104·249	<0·001	Intercept	283·790	<0·001
lg	−0·280	0·011	lg	0·722	<0·001	lg	0·333	<0·001
gr	0·213	0·040				th	0·336	<0·001
th	−0·373	<0·001				sh	0·110	0·078
						th × sh	−0·012	0·003
R^2 = 0·35	RSE = 26·54		R^2 = 0·64	RSE = 17·96		R^2 = 0·40	RSE = 16·01	

population of microorganisms is different (Amon *et al.*, 2007).

Crude fibre

In this study, herbage from all the functional-group monocultures would, based on their mean CF values, fall within an intermediate forage-quality category, according to Peeters (2009) and were within the range of values found by Richter *et al.* (2009) across five semi-natural grasslands typical for European mountainous regions. The results also showed that grass monocultures had higher contents of fibre than, for example, legume species, which confirms the findings on fibre content of different plant species reported by Buxton and Redfearn (1997). Furthermore, the presence of grasses had an effect of increasing the CF across all mixtures, while that of legumes had a decreasing effect (Table 3).

In previous work, NDF (closely correlated with CF) was found to be higher in the high-SR plots than in low-SR plots in the autumn cut of a multi-site

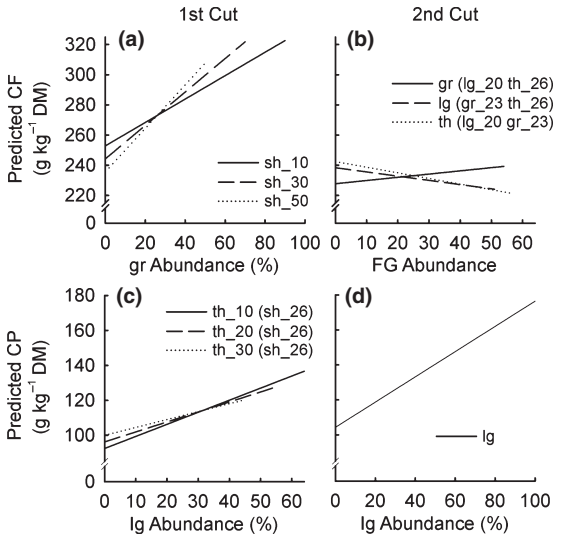

Figure 3 Predictions of crude fibre (CF) and crude protein (CP) according to the multiple regression models in Table 5. (a) Interaction between small herbs (sh) and grasses (gr) affecting predicted CF in the first cut was plotted for three levels of sh abundance (10, 30 and 50%). (b) The three functional groups affecting CF in the second cut were plotted individually along the respective functional-group (FG) abundance, while the other two functional groups were kept constant at their mean value. (c) Interaction between tall herbs (th) and legumes (lg) affecting predicted CP in the first cut was plotted for three levels of th abundance (10, 30 and 50%), while sh was kept constant at its mean value. (d) Lg abundance affecting CP in the second cut.

grassland experiment, whereas no significant effect was found in the summer cut (White *et al.*, 2004). The slope for CF with increasing SR found in this study was positive in both cuts. White *et al.* (2004) attributed both the variation in biomass quality and the variation in SR to differences in nutrient availability in the soil. In the Jena Experiment, the initial variance in soil properties across the experimental site was accounted for in the statistical analysis, and therefore, the positive slope of the SR–CF relation can be attributed to species diversity and its effects on, for instance, soil nutrient availability.

It is often argued that SR effects are due to the so-called sampling effect (e.g. the probability of including more high-yielding species with increasing SR, which then leads to higher biomass). If this were true, it would be reflected, among other factors, in the dominance structure of the plots; however, Khalsa *et al.* (2012) showed no notable change in dominance structure with increasing SR. Furthermore, the independence of SR from functional-group presence/absence could be proven by fitting each functional-group term individually before SR in the hierarchy of the statistical model. It turned out that SR was not affected by any functional-group term except for grasses, which cancelled the significance of SR (Figure 2). Thus, it appears that the existing effect of SR on CF is dependent on the presence of grasses in the grassland mixture. However, this might be partly due to the characteristics of the presence/absence approach for grasses in the Jena Experiment. When the SR relation was plotted for grasses present (+gr) and grasses absent (−gr), it revealed, especially at low-SR levels, an undesired bias (Figure 2). While the mean value over all the one-species plots of −gr contains small-herb, tall-herb and legume monocultures, +gr is solely a mean value of the different grass species sown. It is apparent that the mean of −gr will be lower in CF than the +gr mean, as it contains all the non-grass species, and these are known to have lower CF contents than grasses (Buxton and Redfearn, 1997). In the +gr plots with SR = 2, grasses are no longer in monoculture but are in combination with either of the other functional groups. Thus, differences between +gr and −gr were less pronounced at this SR level and, consequently, they declined further with increasing SR. Thus, it can be assumed that the SR–CF relation at SR > 2 is no longer intermingled with the presence of grasses. In the second cut, however, the 2- and 4-species plots of +gr and −gr have very similar mean values and start to diverge again with increasing SR.

The prediction of CF content from functional-group abundance, which was generated through multiple regression analysis, included an interaction between grass and small-herb abundance in the first cut (Table 5). Plotting this interaction and its effect on predicted CF (CF_{pred}) content revealed that increasing small-herb contents led to a steeper slope for CF_{pred} with increasing grass abundance (Figure 3a). This may reflect stress due to competition between plants of different functional groups but also within a functional group. Competition could occur in the form of small-growing species being forced to grow taller to intercept more light in the presence of more tall-growing species. This in turn would increase fibre contents, as it is known that fibre, and especially lignin, plays an important role in the strength and rigidity of plant cell walls (Buxton and Redfearn, 1997). In the second cut, however, no interactions between functional groups could be observed, and thus, the three functional-group abundances affected CF_{pred} independently (Figure 3b).

Regarding multiple regression analysis on ecological data, Graham (2003) suggested that co-linearity between factors can cause problems with the interpre-

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Table 6 Analysis of Variance of substrate-specific methane yield ($CH_{4\ sub}$), biomass yield and area-specific methane yield ($CH_{4\ area}$) as function of diversity parameters (species richness (SR), functional-groups richness (FGR), functional-group composition (FGC) and presence/absence of individual functional groups. (A) The overall effects of diversity and the significance of cut were tested. The effects of individual functional groups were tested separately for (B) 1st cut and (C) 2nd cut. Arrows indicate increase (↑) or decrease (↓) with presence of respective functional group or with increasing SR.

		$CH_{4\ sub}$ (l_n kg^{-1} VS)		Biomass yield (t ha^{-1})		$CH_{4\ area}$ (m^3 ha^{-1})	
Factor	**DF**	***F***	***P***	***F***	***P***	***F***	***P***
(A) Block	3	1	0·458	3	0·065	2	0·069
Log (SR)	1	31	<0·001↓	31	<0·001↑	26	<0·001↑
FGR	1	1	0·332	2	0·124	2	0·143
FGC	11	9	<0·001	2	0·014	3	0·006
Residuals	59						
Repeated measures							
Cut	1	<1	0·680	192	<0·001	188	<0·001
Cut × Log (SR)	1	3	0·079	10	0·002	7	0·010
Cut × FGR	1	<1	0·943	<1	0·613	<1	0·690
Cut × FGC	11	3	0·010	2	0·045	2	0·120
Residuals	62						
(B) 1st cut							
Block	3	3	0·037	3	0·048	3	0·040
Log (SR)	1	19	<0·001↓	30	<0·001↑	24	<0·001↑
Legumes	1	6	0·016↑	9	0·004↑	13	0·001↑
Grasses	1	29	<0·001↓	1	0·378	<1	0·989
Tall herbs	1	<1	0·789	<1	0·957	<1	0·959
Small herbs	1	9	0·004↑	2	0·129	2	0·218
FGC	8	2	0·021	2	0·091	2	0·164
Residuals	59						
(C) 2nd cut							
Block	3	4	0·010	4	0·015	3	0·027
Log (SR)	1	9	0·004↓	16	<0·001↑	14	0·001↑
Legumes	1	12	0·001↑	17	<0·001↑	21	<0·001↑
Grasses	1	17	<0·001↓	<1	0·826	<1	0·872
Tall herbs	1	6	0·021↑	1	0·471	<1	0·567
Small herbs	1	3	0·088	1	0·389	1	0·297
FGC	8	1	0·311	1	0·668	1	0·679
Residuals	59						

tation of the results if $r > 0{\cdot}28$. As r was on average 0·33 between the functional-group abundances, the results of the multiple regression analysis may not be entirely free of bias. However, the presented regression models help to interpret possible mechanisms driving biomass quality that are reasonable from a biological point of view.

Crude protein

According to Peeters (2009), the functional-group monocultures of grasses would be considered to be of low quality, whereas legume monocultures almost reach high-quality standards. The higher CP content of herbage from legume monocultures relative to grass monocultures is widely known (Posler *et al.*, 1993; Paulson *et al.*, 2008), and this was also supported by the significant functional-group effects (Table 3).

For biogas production, CP is an important constituent, as it allows relatively high specific methane yields (800 l_n kg^{-1} VS) and methane contents in the biogas (70%), as well as high decomposition rates (VDI, 2006). The overall mean CP value in this study was in the range of values found by Richter *et al.* (2009).

Similar to the results of an earlier grassland study with multiple sites (White *et al.*, 2004), CP was lower when SR was high. However, in the study by White *et al.* (2004), nutrient availability was found to be the driver for biomass quality and SR. In other studies,

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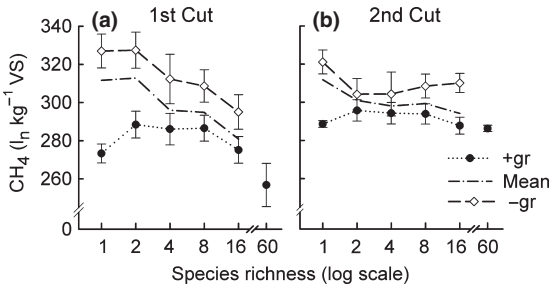

Figure 4 $CH_{4\ sub}$ yields plotted with increasing species richness (SR) and separate for both cuts (a, b). Data were plotted separately for grasses present (+gr: solid circles and dotted line) and grasses absent (−gr: open diamonds and dashed line). Overall trend of mean values with increasing SR was indicated by a dash-dotted line. Values are means of both years. Error bars indicate standard error.

high nutrient inputs led to low SR, with only a few species dominating the vegetation, and low nutrient inputs gave room for the coexistence of a larger pool of species (Vermeer and Berendse, 1983; Mitchley, 2001). In the Jena Experiment, SR was only a part of the experimental design, and as no fertilizers were applied, the differences in nutrient availability could be considered to have occurred in response to the SR level. In an earlier study on the Jena Biomass, it was found that the plant-available N in the soil solution decreased with increasing SR (Oelmann *et al.*, 2007). In other words, nutrient availability was a result of the competition of a certain number of species for the available nutrients. Thus, it was not surprising to find that nitrogen (N = CP/6·25) availability was lower in high-species plots than in plots with a low number of species.

To better understand the impact of SR on CP, it was tested for any dependency on functional-group presence/absence. When legumes were fitted in the model before SR, no changes in the significance of the SR effect were observed. In contrast, fitting grass presence/absence first cancelled its significance (Figure 2). Thus, it appears that the existing SR effect on CP is dependent on the presence of grasses in the grassland mixture. However, this might again be due to the characteristics of the presence/absence of grasses as discussed for CF. Here, the mean value of all the one-species −gr plots is much higher in CP than the +gr plots. This was due to the higher CP concentrations in the herb and legume monocultures as compared to the grass monocultures (Table 4). The legume species may have played an important role as they can supply the system with additional N through atmospheric-N fixation, and high concentrations of N in their tissue are released into the soil during their decomposition (Tomm *et al.*, 1995). In both cuts, the 4-, 8- and 16-species plots of +gr and −gr are clearly different in magnitude but now follow the same pattern. Thus, it can be concluded that the negative SR–CP relation is independent of the presence of grasses at SR > 2.

A significant interaction between tall herbs and legumes occurred for predicted CP (CP_{pred}) in the first cut (Table 5). To visualize this interaction, CP_{pred} was plotted over a range of legume abundances at different levels of tall-herb abundance, while small-herb abundance was kept constant at its mean (Figure 3c). With increasing proportion of tall herbs, the slope of CP_{pred} was lower, which means that the increasing effect of legumes on CP was weakened. This may indicate a lower N-fixation by legumes due to the higher abundance of tall-growing herbs. Regarding the growth of N-fixing species, it is known that shading may limit their growth in relation to non-fixing species, and thus, N-limitation in an ecosystem may be stronger (Vitousek and Field, 1999). This is further supported by the findings that N-fixing plants have difficulties growing through communities with dense canopy structure (Bormann and Gordon, 1984). However, N-fixation by legumes was not measured in this study

Table 7 Substrate-specific ($CH_{4\ sub}$), area-specific ($CH_{4\ area}$) and annual $CH_{4\ sub}$ yield of the different functional-group monocultures and all functional-group mixtures. Values are means of both years.

	Grasses		Legumes		Small herbs		Tall herbs		All FG mixtures	
Methane yield	**Mean**	**s.d.**	**Mean**	**s.d.**	**Mean**	**±s.d.**	**Mean**	**s.d.**	**Mean**	**s.d.**
$CH_{4\ sub}$ (l_n kg^{-1} VS)										
1st cut	270	±8·4	325	±30·7	336	±19·1	304	±27·7	284	±27·5
2nd cut	289	±4·1	321	±16·2	300	±24·5	317	±14·4	296	±14·7
$CH_{4\ area}$ (m^3 ha^{-1})										
1st cut	876	±180	753	±394	608	±394	623	±348	1113	±316
2nd cut	384	±140	565	±372	257	±64	268	±84	562	±207
Annual	1260	±221	1317	±696	865	±413	892	±340	1674	±487

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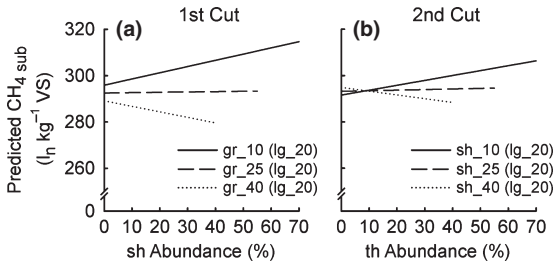

Figure 5 Predictions of substrate-specific methane yields ($CH_{4\ sub}$) according to the multiple regression models in Table 5. (a) Interaction between small herbs (sh) and grasses (gr) affecting predicted $CH_{4\ sub}$ in the first cut plotted for three levels of gr abundance (10, 25 and 40%), while legumes (lg) were kept constant at their mean. (b) Interaction between tall herbs (th) and sh affecting $CH_{4\ sub}$ in the second cut plotted for three levels of sh abundance (10, 25 and 40%), while lg was kept constant at its mean.

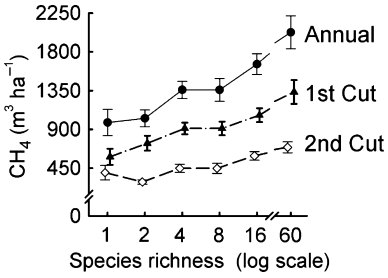

Figure 6 $CH_{4\ area}$ yield, separate for both cuts, and annual $CH_{4\ area}$ yield plotted with increasing species richness. Values are means of both years. Error bars indicate standard error.

and cannot be identified as the driver of the legume–tall herb interaction.

Crude fat and nitrogen-free extract

Regarding anaerobic digestion, CL has the highest biogas yield of all chemical constituents (1390 l_n kg^{-1} VS, according to VDI 4630 (VDI, 2006)) but the slowest conversion rates due to low bioavailability (Weiland, 2010). Richter *et al.* (2009) found a mean CL value that was slightly lower than that found here, yet the grass monoculture value in the first cut was similar to theirs. In general, there was no large variance between the other functional monocultures and the two cuts except for grasses, which had almost double the amount of CL in the second than in the first cut. This may be due to the rather inaccurate prediction of CL by NIRS (R^2 = 0·74, RPD = 1·70). However, estimated $CH_{4\ sub}$ according to VDI 4630 (VDI, 2006) was barely affected by the CL content, especially in the second cut (R^2 = 0·03), which is not surprising, as CL contributed on average to only 2·1% DM of the total biomass.

NfE, even though it contributes on average to 51·9% DM of the total biomass, affects the biogas production as part of the carbohydrate fraction together with CF. The variance between the functional-group monocultures was small, and there was no significant trend with SR.

Substrate-specific methane yields

$CH_{4\ sub}$ was higher than $CH_{4\ sub}$ as estimated by Richter *et al.* (2009) across five different extensive grassland sites, probably due to lower CP values, while CF was similar to the values found here. Gerstl (2008) (also cited in Prochnow *et al.*, 2009) measured $CH_{4\ sub}$ in a batch experiment on extensive grassland silage of 290 l_n kg^{-1} VS in the first cut (June) and 299 l_n kg^{-1} VS in the second cut (October), which were very similar values to the results presented here.

Richter *et al.* (2009) compared values from batch experiments with values calculated according to VDI 4630 (VDI, 2006) and found a close relation (R^2 = 0·99). However, their experiment consisted of only five different plant communities, whereas the data in the study presented here were obtained from a total of 82 different communities. The digestibility data of University of Hohenheim (2007) are limited to two types of grassland communities (grass-rich and grass-poor) and several stages of maturity. Further digestibility values had to be extrapolated to account for all the mixtures occurring in the Jena Experiment (Table 2). This is a potential source of error that would have to be specified by digestibility experiments on the specific plant communities. Nevertheless, the reasonable values obtained from this approach, as compared to values from batch experiments, support this method.

The increasing effect of legume presence on $CH_{4\ sub}$ (Table 6) corresponds to reduced CF and increased CP contents (Table 3). The opposite is true for grasses, which increased the CF and decreased the CP and thereby affected $CH_{4\ sub}$ negatively.

The decrease in $CH_{4\ sub}$ with increasing SR is in line with the patterns of CF and CP values changing along the SR gradient and confirms the hypothesis stated in the Introduction.

To test whether the relationship of $CH_{4\ sub}$ and SR was dependent on any functional-group presence/absence, the sequence of model fitting was changed. No functional group fitted before SR in the first cut had any effect on its significance, while fitting grasses before SR in the second cut cancelled the significance of the SR effect. This can be explained following the same line of argument as for CF and CP. The slope of

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+gr and −gr in the first cut of the high-SR plots (>2) is very similar, and the SR–$CH_{4\ sub}$ relation can be considered a true SR effect at SR > 2 (Figure 4). In the second cut, the slopes at high SR kept diverging, and the SR effect was dependent on the presence/absence of grasses over the entire range of SR (Figure 4). Taking a closer look at the second cut, it appears that, in the absence of grasses, increasing the SR has a slight effect of increasing the $CH_{4\ sub}$ (at SR > 1). This is in contrast to the first cut and may result from the slight decline in CF with increasing SR in the −gr plots of the second cut (Figure 2b).

Multiple regression analysis revealed interactions between functional-group abundances that may further shed light on the relationships between species composition and $CH_{4\ sub}$ (Figure 5). In the first cut, the strong increasing effect of small herbs on predicted $CH_{4\ sub}$ was only noticeable when grass abundance was low. It is possible that the imposed competition on small herbs with increasing grass abundance caused a decline in $CH_{4\ sub}$. Thus, as soon as the proportion of grasses present exceeded 10%, the small herbs had to grow taller to intercept more light and thereby strengthen their vertical growth through fibre and possibly lignin structures (Buxton and Redfearn, 1997). In the second cut, a similar interaction could be observed between small and tall herbs. According to the multiple regression analysis, tall herbs had a highly significant and increasing effect on predicted $CH_{4\ sub}$, while small herbs, as main effect, had no significant effect. Yet, with increasing abundance of small herbs, the increasing effect of tall herbs ceased, and the reason for this may be the due to same competition mechanisms as described above.

Area-specific methane yields

The wide range of values among the biomass yield and $CH_{4\ area}$ of all plots was due to the fact that especially the low species-number communities were kept off equilibrium through weeding, and thus, they had a low biomass, while the sixty-species plots could utilize most of the available ecological niches for biomass production. The advantages and disadvantages of such an approach were discussed in Khalsa *et al.* (2012).

The differences in $CH_{4\ area}$ between the two cuts were comparable with the results presented by Amon *et al.* (2007) for a two-cut hill site where the first cut contributed about 65% to the annual $CH_{4\ area}$, while the second cut only contributed 35%. However, in their study, the annual biomass yield was higher (6·4 t DM a^{-1}), while $CH_{4\ sub}$ was considerably lower (first cut: 171 l_n kg^{-1} VS; second cut: 128 l_n kg^{-1} VS) than the yields in the this study.

As revealed by the statistical analysis (Table 6), legumes had a strong effect on $CH_{4\ area}$. This was partly due to the increasing $CH_{4\ sub}$ but much more so due to the increasing biomass yield. It can be assumed that this was not simply a sampling effect but rather a combination of facilitating plant growth of other species and promoting complementary resource use among species, as discussed in Khalsa *et al.* (2012).

Estimated mean annual $CH_{4\ area}$ was similar to data obtained by Gerstl (2008), who measured biomass from a two-cut Alpine Hill site. The strong positive slope of the SR–annual $CH_{4\ area}$ relationship was mainly driven by the SR–biomass relationship. The first cut especially showed a strong SR–$CH_{4\ area}$ relation, whereas it was less pronounced in the second cut (Figure 4). The negative SR–$CH_{4\ sub}$ slope was thus compensated by the positive SR–biomass slope.

Focusing on the most relevant SR levels in the extensive grasslands (8–6–60 species), annual $CH_{4\ area}$ still increased by 669 m^3 ha^{-1} a^{-1} along a species gradient from an eight- to sixty-species plot and by 370 m^3 ha^{-1} a^{-1} from a sixteen- to a sixty-species plot. The range found in this study from an eight-species plot (1358 m^3 ha^{-1} a^{-1}) to a sixty-species plot (2027 m^3 ha^{-1} a^{-1}) is comparable with values obtained by Gerstl (2008), measured for a two-cut hill site (1580 m^3 ha^{-1} a^{-1}) and a four-cut valley site (2359 m^3 ha^{-1} a^{-1}).

Conclusion

Chemical constituents most relevant for methane production from anaerobic digestion were successfully investigated along experimental diversity gradients. The results provide a novel insight into plant diversity –bioenergy relations. Methane yields could be derived from the concentrations of chemical constituents and reported DM digestibility, and these yield measures could be used to express the bioenergy potential of this type of experimental grassland biomass, which is a type that could occur in extensive grassland managed for nature conservation purposes.

Several effects of SR on digestion-relevant parameters were found: (i) CP decreased with increasing SR, while CF increased; however, these two effects were not fully disentangled from the presence of grasses, (ii) increasing SR resulted in lower $CH_{4\ sub}$ and (iii) increasing SR increased $CH_{4\ area}$. Functional groups, especially legumes and grasses, played key roles in the variation in herbage chemical constituents and $CH_{4\ sub}$ respectively.

The results of this study suggest that for a two-cut extensive management, it is important to maintain legume abundance, as legumes affect $CH_{4\ sub}$ directly and affect $CH_{4\ area}$ through positive feedback on biomass yield.

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Acknowledgments

The authors thank the German Research Foundation (DFG) for funding the project (WA 2135/2-1) and the technical staff of the Department of Grassland Science and Renewable Plant Resources for their support.

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